1. A pair of dice is thrown 5 times. For each throw, a total of 5 is considered a success. If the probability of at least 4 successes is $\frac{k}{3^{11}}$, then k is equal to

[2023 (06 Apr Shift 1)]

(1) 82
(2) 75
(3) 164
(4) 123

2. Three dice are rolled. If the probability of getting different numbers on the three dice is $\frac{p}{q}$, where p and q are co-prime, then $q-p$ is equal to

[2023 (06 Apr Shift 2)]

(1) 2
(2) 1
(3) 3
(4) 4

3. In a bolt factory, machines A, B and C manufacture respectively 20%, 30% and 50% of the total bolts. Of their output 3, 4 and 2 percent are respectively defective bolts. A bolt is drawn at random from the product. If the bolt drawn is found the defective then the probability that it is manufactured by the machine C is

[2023 (08 Apr Shift 1)]

(1) $\frac{5}{14}$
(2) $\frac{9}{28}$
(3) $\frac{3}{7}$
(4) $\frac{2}{7}$

4. If the probability that the random variable X takes values x is given by $P(X=x)=k(x+1)3^{-x}$, $x=0,1,2,3,\ldots\ldots$, where k is a constant, then $P(X\geq 2)$ is equal to

[2023 (08 Apr Shift 2)]

(1) $\frac{7}{27}$
(2) $\frac{7}{18}$
(3) $\frac{11}{18}$
(4) $\frac{20}{27}$

5. Let N denote the sum of the numbers obtained when two dice are rolled. If the probability that $2^N < N!$ is $\frac{m}{n}$ where m and n are coprime, then $4m-3n$ is equal to

[2023 (10 Apr Shift 1)]

(1) 6
(2) 12
(3) 10
(4) 8

6. Let a die be rolled n times. Let the probability of getting odd numbers seven times be equal to the probability of getting odd numbers nine times. If the probability of getting even numbers twice is $\frac{k}{2^{15}}$, then k is equal to

[2023 (10 Apr Shift 2)]

(1) 60
(2) 15
(3) 90
(4) 30

7. Let $S=\{M=[a_{ij}],\ a_{ij}\in\{0,1,2\},\ \{1\leq i,j\leq 2\}\}$ be a sample space and $A\{M\in S : M \text{ is invertible}\}$ be an even. Then $P(A)$ is equal to

[2023 (11 Apr Shift 1)]

(1) $\frac{16}{27}$
(2) $\frac{47}{81}$
(3) $\frac{49}{81}$
(4) $\frac{50}{81}$

8. Let the probability of getting head for a biased coin be $\frac{1}{4}$. It is tossed repeatedly until a head appears. Let N be the number of tosses required. If the probability that the equation $64x^2+5Nx+1=0$ has no real root is $\frac{p}{q}$, where p and q are co-prime, then $q-p$ is equal to..........

[2023 (11 Apr Shift 2)]

9. Two dice A and B are rolled. Let the numbers obtained on A and B be α and β respectively. If the variance of $\alpha-\beta$ is $\frac{p}{q}$, where p and q are co-prime, then the sum of the positive divisors of p is equal to

[2023 (12 Apr Shift 1)]

(1) 72
(2) 36
(3) 48
(4) 31

10. A fair n $(n>1)$ faces die is rolled repeatedly until a number less than n appears. If the mean of the number of tosses required is $\frac{n}{9}$, then n is equal to

[2023 (12 Apr Shift 1)]

11. A coin is biased so that the head is 3 times as likely to occur as tail. This coin is tossed until a head or three tails occur. If X denotes the number of tosses of the coin, then the mean of X is

[2023 (13 Apr Shift 1)]

(1) $\frac{38}{16}$
(2) $\frac{15}{16}$
(3) $\frac{21}{16}$
(4) $\frac{81}{64}$

12. The random variable X follows binomial distribution $B(n,\ p)$, for which the difference of the mean and the variance is 1.
If $2\ P(X=2)=3\ P(X=1)$, then $n^2 P\big(X>1\big)$ is equal to

[2023 (13 Apr Shift 2)]

(1) 15
(2) 11
(3) 12
(4) 16

13. A bag contains 6 white and 4 black balls. A die is rolled once and the number of balls equal to the number obtained on the die are drawn from the bag at random. The probability that all the balls drawn are white is

[2023 (15 Apr Shift 1)]

(1) $\frac{1}{4}$
(2) $\frac{11}{50}$
(3) $\frac{1}{5}$
(4) $\frac{9}{50}$

ANSWER KEYS

1. (4)	**2.** (4)	**3.** (1)	**4.** (1)	**5.** (4)	**6.** (1)	**7.** (4)	**8.** (27)
9. (3)	**10.** (10)	**11.** (3)	**12.** (2)	**13.** (3)			

1. (4)

Given that,

A pair of dice is thrown 5 times.

Required Sample space is $\{(1,1),(1,2),(1,3)\ldots\ldots\ldots.(6,6)\}$

Total number of observations $= 36$

Now, a sum of 5 is observed in $\{(1,4),(4,1),(2,3),(3,2)\}$

No of favourable outcomes $= 4$

$P(\text{success})\ p = \frac{4}{36} = \frac{1}{9}$

$P(\text{failure})\ q = 1 - \frac{1}{9} = \frac{8}{9}$

We know that for a binomial distribution, $B(n, p)$

$P(X = k) = {}^nC_k p^k q^{n-k}$ where $k \geq 0$ and $p + q = 1$

$\therefore$Required probability

$\Rightarrow P(X \geq 4) = P(X = 4) + P(X = 5)$

$= {}^5C_4\left(\frac{1}{9}\right)^4 \frac{8}{9} + {}^5C_5\left(\frac{1}{9}\right)^5$

$= 5 \cdot \frac{8}{9^5} + \frac{1}{9^5} = \frac{41}{9^5}$

$= \frac{41 \times 3}{3^{10} \times 3} = \frac{123}{3^{11}}$

But given that $P(X \geq 4) = \frac{k}{3^{11}}$

$\Rightarrow \frac{k}{3^{11}} = \frac{123}{3^{11}}$

$\Rightarrow k = 123$

Hence, this is the correct option.

2. (4)

Given that three dice are thrown.

The total number of outcomes when three dice are thrown together is $6^3 = 6 \times 6 \times 6$.

The number of outcomes such that all the outcomes are different is $= 6 \times 5 \times 4$.

For ex: If the outcome in dice 1 is $''6''$ then the number of outcomes for dice 2 should be $5\ (1,2,3,4,5)$ which excludes the outcome $''6''$.

Hence, the required probability is $= \frac{6 \times 5 \times 4}{6 \times 6 \times 6}$

$= \frac{20}{36} = \frac{5}{9} = \frac{p}{q}$

$\Rightarrow q - p = 9 - 5 = 4$

Therefore, the required answer is 4.

3. (1)

Let $X \equiv$Event that product is defective, then

$P\left(\frac{X}{A}\right) = \frac{3}{100}$

$P\left(\frac{X}{B}\right) = \frac{4}{100}$

$P\left(\frac{X}{C}\right) = \frac{2}{100}$

Now,

$$P\left(\frac{C}{X}\right) = \frac{P(C)P\left(\frac{X}{C}\right)}{P(A)P\left(\frac{X}{A}\right) + P(B)P\left(\frac{X}{B}\right) + P(C)P\left(\frac{X}{C}\right)}$$

$$\Rightarrow P\left(\frac{C}{X}\right) = \frac{\frac{50}{100} \times \frac{2}{100}}{\frac{20}{100} \times \frac{3}{100} + \frac{30}{100} \times \frac{4}{100} + \frac{50}{100} \times \frac{2}{100}}$$

$\Rightarrow P\left(\frac{C}{X}\right) = \frac{100}{60 + 120 + 100}$

$\Rightarrow P\left(\frac{C}{X}\right) = \frac{5}{14}$

Hence this is the correct option.

4. (1)

Given,

The probability that the random variable X takes values x is given by $P(X=x)=k(x+1)3^{-x}$, $x=0,1,2,3,\ldots\ldots$, where k is a constant,

Now we know that,

$P(X=0)+P(x=1)+P(x=2)+\ldots\ldots\ldots=1$

$\Rightarrow \frac{k}{3^0}+\frac{2k}{3^1}+\frac{3k}{3^2}+\ldots\ldots=1$

$\Rightarrow k\left(1+\frac{2}{3}+\frac{3}{3^2}+\ldots\right)=1$

Now finding,

$S=1+\frac{2}{3}+\frac{3}{3^2}+\frac{4}{3^3}+\ldots \quad \ldots(1)$

$\Rightarrow \frac{S}{3}=\frac{1}{3}+\frac{2}{3^2}+\frac{3}{3^2}+\ldots\ldots \quad \ldots(2)$

Now on subtracting above two equation we get,

$\Rightarrow \frac{2S}{3}=1+\frac{1}{3}+\frac{1}{3^2}+\ldots\ldots$

$\Rightarrow S=\frac{9}{4}$

Hence, $k\left(1+\frac{2}{3}+\frac{3}{3^2}+\ldots\right)=1$

$\Rightarrow k=\frac{4}{9}$

Now finding,

$P(X\geq 2)=P(2)+P(3)+\ldots\ldots$

$\Rightarrow P(X\geq 2)=1-P(0)-P(1)$

$\Rightarrow P\left(X\geq 2\right)=1-\left(\frac{k}{1}+\frac{2k}{3}\right)=1-\frac{20}{27}=\frac{7}{27}$

5. (4)

Given that,

$2^N<N!$

N is the sum of numbers of two dice.

$\Rightarrow 2\leq N\leq 12$

Let us check the given condition $2^N<N!$

$N=1$ (not possible) $\rightarrow 0$

$N=2$ (not possible) $\rightarrow 1$

$N=3$ (not possible) $\rightarrow 2$

$N=4$ (possible)

We need to find the probability for $N\geq 4$.

$\Rightarrow P(N\geq 4)=1-P(N=2)-P(N=3)$

$=1-\frac{1}{36}-\frac{2}{36}$

$=\frac{11}{12}=\frac{m}{n}$

Now let us find $4m-3n$.

$=4\times 11-3\times 12=8$.

Hence, the required answer is 8.

6. (1)

Given that $P(\text{getting odd 7 times})=P(\text{getting odd 9 times})$.

This is the case of binomial distribution with n successes and probability of success $=p=\frac{3}{6}$ and failure is $q=1-p=\frac{1}{2}$.

We know that $P(X=k)={}^nC_k(p)^k(q)^{n-k}$

Now $P(X=7)=P(X=9)$

$\Rightarrow {}^nC_7\left(\frac{1}{2}\right)^7\left(\frac{1}{2}\right)^{n-7}={}^nC_9\left(\frac{1}{2}\right)^9\left(\frac{1}{2}\right)^{n-9}$

$\Rightarrow {}^nC_7={}^nC_9$

$\Rightarrow {}^nC_{n-7}={}^nC_9$

$\Rightarrow n=16$

$\Rightarrow P\left(\text{2 times even}\right)={}^{16}C_2\left(\frac{1}{2}\right)^{14}\left(\frac{1}{2}\right)^2$

$=\frac{15\times 4}{2^{15}}=\frac{60}{2^{15}}=\frac{k}{2^{15}}$

$\Rightarrow k=60$

Hence this is the required option.

7. **(4)**

Given,

$S=\{M=[a_{ij}],\ a_{ij}\in\{0,1,2\},\ \{1\le i,j\le 2\}\}$

So, let $M=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ where $a,b,c,d\in\{0,1,2\}$

So, total sample space will be $3^4=81$

Now we will find $P(\bar{A})$ or possibilities for $|M|=0$

So, for $|M|=0$ cases will be,

1. $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ or $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ or $\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}\to$ Total matrix $=3$
2. Two $1's$ and Two $0's\to$ Total matrix $=4$
3. Two $2's$ and Two $0's\to$ Total matrix $=4$
4. Two $1's$ and Two $2's\to$ Total matrix$=4$
5. One 1 and three $0\ 's\to$ Total matrix $=4$
6. One 2 and Three $0's\to$ Total matrix $=4$
7. One 1 and one 2 and two $0's\to$ Total matrix $=8$

So, total cases for which $|M|=0$ is 31,

So probability will be $P(\bar{A})=\frac{31}{81}$

Hence, $P(A)=1-\frac{31}{81}=\frac{50}{81}$

8. **(27)**

Given a quadratic equation $64x^2+5nx+1=0$ has no real roots, so

$D<0$

$\Rightarrow 25N^2-4\times 64<0$

$\Rightarrow N<\frac{16}{5}\Rightarrow N=1,2,3$

And, For a biased coin, the probability of getting head is $P=\frac{1}{4}$, and probability of getting tail will be $Q=\frac{3}{4}$

So, required probability is

$\frac{1}{4}+\frac{3}{4}\cdot\frac{1}{4}+\left(\frac{3}{4}\right)^2\cdot\frac{1}{4}+\left(\frac{3}{4}\right)^3\cdot\frac{1}{4}=\frac{p}{q}$

$\Rightarrow \frac{\frac{1}{4}\left(1-\left(\frac{3}{4}\right)^3\right)}{1-\left(\frac{3}{4}\right)}=\frac{p}{q}$

$\Rightarrow 1-\left(\frac{3}{4}\right)^3=\frac{p}{q}$

$\Rightarrow 1-\frac{27}{64}=\frac{p}{q}$

$\Rightarrow \frac{p}{q}=\frac{37}{64}$

$q-p=27$

9. **(3)**

Given that variance of $\alpha-\beta$ is $\frac{p}{q}$

$\alpha \in\{1,\ 2,\ 3,\ 4,\ 5,\ 6\}$

$\beta \in\{1,\ 2,\ 3,\ 4,\ 5,\ 6\}$

$(\alpha-\beta)=0$ (6 case)

$(\alpha-\beta)=-1$ (5 case)

$(\alpha-\beta)=-2$ (4 case)

$(\alpha-\beta)=-3$ (3 case)

$(\alpha-\beta)=-4$ (2 case)

$(\alpha-\beta)=-5$ (1 case)

$(\alpha-\beta)=1$ (5 case)

$(\alpha-\beta)=2$ (4 case)

$(\alpha-\beta)=3$ (3 case)

$(\alpha-\beta)=4$ (2 case)

$(\alpha-\beta)=5$ (1 case)

Mean $=0$

We know that variance $=\frac{\sum_{i=1}^{n} f_i\left(x_i-\bar{x}\right)^2}{\sum_{i=1}^{n} f_i}$

Variance$=\sigma^2=\frac{0^2\times6+2\times1^2\times5+2\times2^2\times4+2\times3^2\times3+2\times4^2\times2\times5^2\times1}{36}$

$=\frac{2}{36}\times(5+16+27+32+25)=\frac{105}{18}=\frac{35}{6}$

$\therefore\ p=35$

Sum of divisors of $p=1+5+7+35=48$

Option (3) is correct.

10. **(10)**

Given,

A fair n $(n>1)$ faces die is rolled repeatedly until a number less than n appears,

x_i	1	2	3	4	$\ldots$
p_i	$\frac{n-1}{n}$	$\frac{1}{n}\cdot\left(\frac{n-1}{n}\right)$	$\frac{1}{n^2}\cdot\frac{n-1}{n}$	$\frac{1}{n^3}\cdot\left(\frac{n-1}{n}\right)$	$\ldots$

Now we know that, mean is given by,

Mean $=\sum_{i=1}^{\infty} p_i x_i=1\cdot\frac{n-1}{n}+\frac{2}{n}\cdot\left(\frac{n-1}{n}\right)+\frac{3}{n^2}\left(\frac{n-1}{n}\right)+\ldots$

$\Rightarrow\frac{n}{9}=\left(1-\frac{1}{n}\right)S\ldots\ldots..(1)$

Where, $S=1+\frac{2}{n}+\frac{3}{n^2}+\frac{4}{n^3}+\ldots\quad\ldots\ldots(2)$

$\frac{1}{n}S=\frac{1}{n}+\frac{2}{n^2}+\frac{3}{n^3}+\ldots\ldots\ldots(3)$

Now subtracting equation $(2)-(3)$ we get,

$\left(1-\frac{1}{n}\right)S=1+\frac{1}{n}+\frac{1}{n^2}+\frac{1}{n^3}+\ldots$

$\Rightarrow\left(1-\frac{1}{n}\right)S=\frac{1}{1-\frac{1}{n}}$

Now putting the value of S in equation (1) we get,

$\Rightarrow\frac{n}{9}=\left(1-\frac{1}{n}\right)\times\frac{1}{\left(1-\frac{1}{n}\right)^2}=\frac{n}{n-1}$

$\Rightarrow n=10$

11. **(3)**

It is given that $P(H)=3P(T)$

$\therefore P\left(H\right)=\frac{3}{4},P\left(T\right)=\frac{1}{4}$

Since coin is tossed till either $1H$ or $3T$ occurs. So, process will end in the maximum of 3 throws.

Xi	1	2	3
$P(Xi)$	$\frac{3}{4}$	$\frac{1}{4}\cdot\frac{3}{4}$	$\frac{1}{4}\times\frac{1}{4}\times\frac{1}{4}+\frac{1}{4}\times\frac{1}{4}\times\frac{3}{4}$

Now we know that,

Mean is given by $\sum_{i=1}^{n} X_i P(X_i)$

Mean$=1\times\frac{3}{4}+2\times\frac{3}{16}+3\times\frac{1}{16}=\frac{21}{16}$

Hence, this is the correct option.

12. (2)

Given that the difference between the mean and variance is 1.

$\Rightarrow np - npq = 1$
$\Rightarrow np(1-q) = 1$
$\Rightarrow np^2 = 1$
Also given that,
$\Rightarrow 2P(X=2) = 3P(X=1)$
$\Rightarrow 2 \cdot {}^nC_2 p^2 q^{n-2} = 3 \cdot {}^nC_1 p \cdot q^{n-1}$
$\Rightarrow 2 \cdot \frac{n \cdot (n-1)}{2} \cdot p = 3 \cdot n \cdot q$
$\Rightarrow (n-1)p = 3(1-p)$
$\Rightarrow \left(\frac{1}{p^2} - 1\right)p = 3\left(1-p\right)$
$\Rightarrow \frac{(1-p)(1+p)}{p} = 3\left(1-p\right)$
$\Rightarrow 1 + p = 3p$
$\Rightarrow \ p = \frac{1}{2}$
$\therefore n = 4.$
Now,
$n^2 P(x > 1) = n^2\left(1 - P(x=1) - P(x=0)\right) = 16\left(1 - {}^4C_1 \cdot \left(\frac{1}{2}\right)^4 - \left(\frac{1}{2}\right)^4\right) = 11$

Hence this is the correct option.

13. (3)

Given,
Bag have 6 white and 4 black balls
Now also given die is rolled and the number denotes the number of ball drawn from the bag,
So, probability all drawn balls are white will be,
$= \frac{1}{6}\left[\frac{{}^6C_1}{{}^{10}C_1} + \frac{{}^6C_2}{{}^{10}C_2} + \frac{{}^6C_3}{{}^{10}C_3} + \frac{{}^6C_4}{{}^{10}C_4} + \frac{{}^6C_5}{{}^{10}C_5} + \frac{{}^6C_6}{{}^{10}C_6}\right]$
$= \frac{1}{6}\left[\frac{6}{10} + \frac{15}{45} + \frac{20}{120} + \frac{15}{210} + \frac{6}{252} + \frac{1}{210}\right]$
$= \frac{504}{2520} = \frac{1}{5}$